Weather Monitoring and Trend Prediction System

A CAPSTONE PROJECT REPORT

Submitted in the partial fulfilment for the Course of

CSA0508-Database Management Systems for Data Security

to the award of the degree of

BACHELOR OF ENGINEERING

IN

Computer Science and Engineering

Submitted by

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Under the Supervision of

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DECLARATION

We, **Deepak. R (192511008.) Vijay. A (192511030)** of the **Computer Science and Engineering,** Saveetha Institute of Medical and Technical Sciences, Saveetha University, Chennai, hereby declare that the Capstone Project Work entitled '**Weather Monitoring and Trend Prediction System'**is the result of our own bonafide efforts. To the best of our knowledge, the work presented here in is original, accurate, and has been carried out in accordance with principles of engineering ethics.

Place:

Date:

Signature of the Students with Names

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BONAFIDE CERTIFICATE

This is to certify that the Capstone Project entitled "**Weather Monitoring and Trend Prediction System**" has been carried out by **Deepak. R (192511008.) Vijay. A (192511030)** under the supervision of **Dr Pramila and Dr Nandhini** and is submitted in partial fulfilment of the requirements for the current semester of the B.E **Computer Science and Engineering** program at Saveetha Institute of Medical and Technical Sciences, Chennai.

SIGNATURE

SIGNATURE

Submitted for the Project work Viva-Voce held on ___________________________________

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ABSTRACT

In recent years, weather prediction systems have become essential tools for supporting daily activities such as agriculture planning, disaster management, transportation, and public safety. Accurate weather forecasting requires the collection, storage, and analysis of large volumes of meteorological data generated continuously from various sources. Managing this data efficiently demands a well-structured and reliable database management system. This project presents the design and implementation of a Weather Prediction System with a strong emphasis on database design and data integrity.

The proposed system collects real-time and historical weather data such as temperature and date-wise observations from external weather APIs. The collected data is stored in a relational database using MySQL, ensuring consistency, accuracy, and minimal redundancy through normalization techniques. The database design includes tables for storing actual weather data and predicted weather values, connected using primary and unique key constraints to maintain data integrity. Structured Query Language (SQL) is used to perform efficient data insertion, retrieval, and updates.

A simple prediction mechanism is implemented using recent historical temperature values to estimate future temperatures. The predicted data is also stored in the database, allowing comparative analysis between actual and predicted values. Visualization techniques using line graphs and bar graphs are employed to present weather trends clearly to users. Additionally, the system provides real-time weather information for the current day through a user-friendly graphical interface.

Overall, this project demonstrates how database management concepts such as relational schema design, normalization, integrity constraints, and query optimization can be effectively applied to a real-world weather prediction application. The system ensures efficient data handling, supports analytical operations, and provides meaningful insights through visualization, making it a practical and scalable solution.

CHAPTER – 1

INTRODUCTION

Weather plays a vital role in influencing human activities, economic planning, and environmental decision-making. Accurate and timely weather information is essential for sectors such as agriculture, aviation, disaster management, and urban planning. With advancements in technology and the availability of real-time meteorological data, weather prediction systems have evolved into data-driven applications that rely heavily on efficient data storage and processing mechanisms.

A Weather Prediction System is an application designed to collect, store, analyze, and forecast weather conditions based on historical and real-time data. At the core of such a system lies a well-structured database that manages weather observations in an organized manner. A robust database ensures that weather data is stored securely, retrieved efficiently, and updated accurately without redundancy or inconsistency.

This project focuses on designing a database-driven weather prediction system using relational database principles. The system retrieves temperature data from an external weather API and stores it in a MySQL database. Using structured queries and proper constraints, the database maintains data integrity while supporting prediction and visualization functionalities. The system also provides a graphical user interface to improve usability and accessibility.

The objective of this project is to demonstrate the application of database management system concepts in building a real-world weather prediction application. It highlights how effective database design, normalization, and query optimization contribute to accurate forecasting, efficient data handling, and meaningful visualization of weather trends.

CHAPTER – 2

PROBLEM IDENTIFICATION AND ANALYSIS

With the increasing availability of real-time weather data, one of the major challenges lies in efficiently managing large volumes of meteorological information. Weather data is generated daily and continuously updated, making it necessary to store historical records while ensuring fast retrieval for analysis and prediction. Without a properly designed database, weather systems may suffer from data redundancy, inconsistency, and slow query performance.

A common problem in basic weather applications is the lack of structured data storage. Many systems rely solely on real-time API responses without maintaining historical records, which limits analysis and forecasting capabilities. Additionally, repeated data insertion without proper constraints can lead to duplicate entries, compromising data integrity and reliability.

Another challenge is integrating prediction results with historical data in a meaningful way. Without a normalized database structure, storing predicted values alongside actual data becomes complex and error-prone. Moreover, users often require visual representations of weather trends, which demand efficient querying and well-organized datasets.

The identified problem can therefore be summarized as the need for a structured, reliable, and scalable database system that can store historical weather data, support prediction mechanisms, prevent data duplication, and enable effective visualization. Addressing these issues requires a database-centric solution that adheres to fundamental DBMS principles.

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SOLUTION DESIGN AND IMPLEMENTATION

The proposed Weather Prediction System is designed using a database-first approach, where data integrity and efficient storage are given priority. The system architecture consists of three major components: data collection, database management, and data visualization.

Database Design

The relational database is designed using MySQL and includes the following core tables:

- **Weather_Data**: Stores historical weather data such as date, temperature, and city name.

```
mysql> select * from weather_data;
+----+------------+-------------+---------+
| id | date       | temperature | city    |
+----+------------+-------------+---------+
|  1 | 2025-12-31 |        25.6 | chennai |
|  2 | 2026-01-01 |        25.6 | chennai |
|  3 | 2026-01-02 |        25.8 | chennai |
|  4 | 2026-01-03 |        25.6 | chennai |
|  5 | 2026-01-04 |        25.1 | chennai |
|  6 | 2026-01-05 |        24.8 | chennai |
+----+------------+-------------+---------+
6 rows in set (0.03 sec)
```

- **Predicted_Weather**: Stores predicted temperature values along with corresponding future dates and city information.

```
mysql> select * from predicted_weather;
+----+------------+-------------+---------+
| id | date       | temperature | city    |
+----+------------+-------------+---------+
|  1 | 2026-01-06 |     25.4667 | chennai |
|  2 | 2026-01-07 |     25.7667 | chennai |
|  3 | 2026-01-08 |     26.0667 | chennai |
+----+------------+-------------+---------+
3 rows in set (0.03 sec)
```

Primary keys and unique constraints are applied to prevent duplicate entries for the same date and city combination. The database schema is normalized to eliminate redundancy and ensure consistency.

Data Collection and Storage

Weather data is collected from an external weather API based on the user-provided city name. The retrieved data is processed and inserted into the database using SQL queries. Duplicate data is handled using constraints and conditional insertion logic to maintain integrity.

Prediction Mechanism

A simple prediction algorithm is implemented by calculating the average temperature of recent historical data and incrementally estimating future temperatures. These predicted values are stored in the database for further analysis and visualization.

Visualization and User Interface

The system provides a graphical user interface using Python’s Tkinter library. Line graphs and bar graphs are generated using Matplotlib to display actual and predicted temperature trends. A separate feature allows users to view the current day’s weather conditions through a dedicated button.

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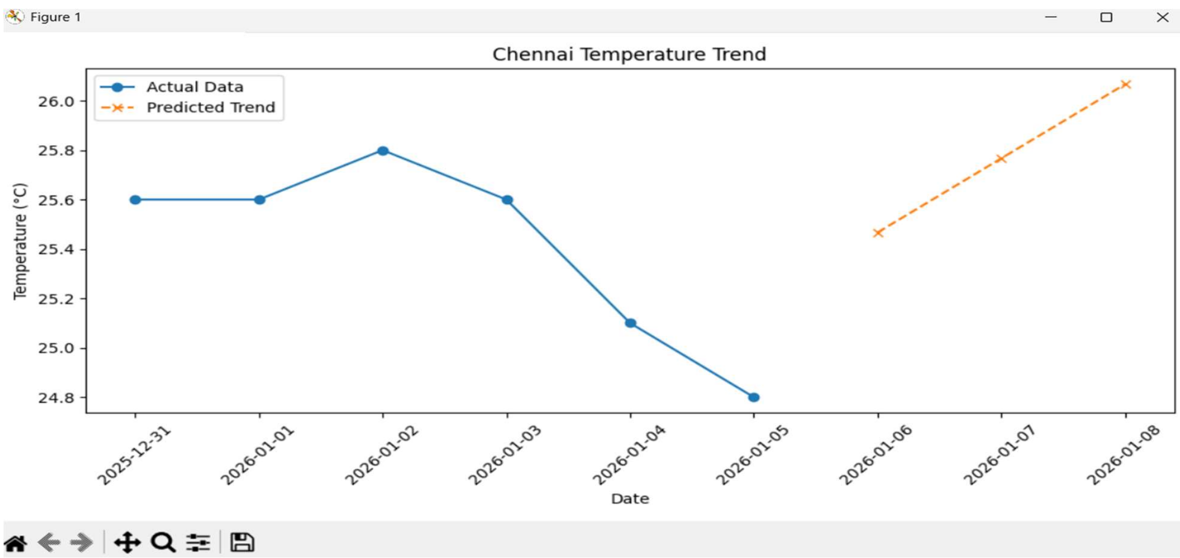
RESULTS

The Weather Prediction System successfully stores historical and predicted weather data in the database without redundancy. SQL queries efficiently retrieve data for visualization and analysis. The line graph clearly represents temperature trends over time, while the bar graph allows comparison between actual and predicted values.

The current weather feature accurately displays real-time temperature data, enhancing the usability of the system. The database structure ensures fast query execution and reliable data storage even when the system is executed multiple times.



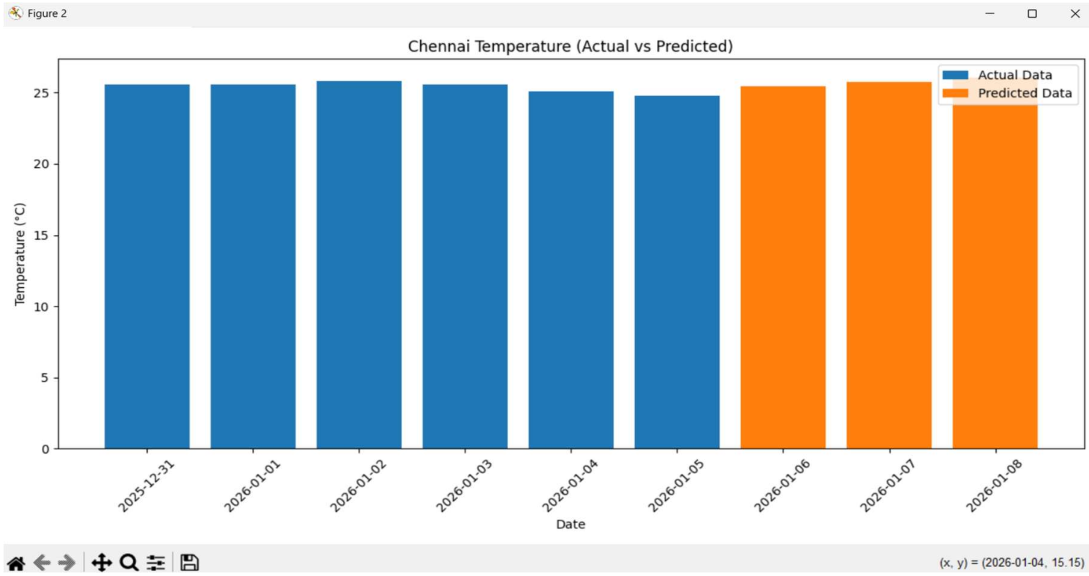


Figure 2
Chennai Temperature (Actual vs Predicted)
Actual Data
Predicted Data
Temperature (°C)
0
5
10
15
20
25
2025-12-31
2026-01-01
2026-01-02
2026-01-03
2026-01-04
2026-01-05
2026-01-06
2026-01-07
2026-01-08
Date
(x, y) = (2026-01-04, 15.15)

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REFLECTION ON LEARNING AND PERSONAL DEVELOPMENT

Working on the Weather Prediction System was a valuable learning experience that strengthened my understanding of Database Management System concepts and their practical applications. This project helped me move beyond theoretical knowledge and apply database principles in a real-world scenario involving real-time data.

Through this project, I gained hands-on experience in designing relational database schemas, applying normalization techniques, and maintaining data integrity using primary and unique key constraints. Writing and executing SQL queries for data insertion, retrieval, and updates improved my understanding of how databases handle large and dynamic datasets efficiently.

The integration of historical weather data with prediction logic enhanced my analytical thinking and problem-solving skills. I also learned how databases can support analytical operations and visualizations, rather than serving only as data storage systems. Designing a user-friendly interface further improved my understanding of how database-driven applications interact with users.

Overall, this project contributed significantly to my technical skills, confidence, and understanding of DBMS concepts. It also improved my ability to design structured, reliable, and scalable data-centric applications, which will be beneficial for future academic and professional work.

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CONCLUSION

The Weather Prediction System developed in this project demonstrates the effective application of Database Management System concepts in a real-world scenario. The system efficiently stores and manages historical and real-time weather data using a relational database, ensuring data integrity and consistency through proper normalization and constraints.

By integrating weather data from external APIs, the system allows structured storage and retrieval of temperature records for analysis and prediction. The use of SQL queries enables efficient data manipulation, while unique key constraints prevent data duplication. A simple prediction mechanism based on historical data is implemented to estimate future temperatures, and the results are stored in the database for comparison.

The system also provides clear visualization of weather trends using graphical representations, which helps users easily understand temperature variations over time. The inclusion of a current weather feature enhances usability by providing real-time information.

Overall, this project highlights the importance of database design, normalization, and query optimization in developing reliable and scalable applications. The Weather Prediction System successfully meets its objectives and serves as a strong foundation for future enhancements using advanced forecasting techniques.

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